



Factorization-2

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Factorization using Identities.

$$\begin{aligned}
 &\rightarrow (a+b)^2 = a^2 + b^2 + 2ab \\
 &\rightarrow (a-b)^2 = a^2 + b^2 - 2ab \\
 &\rightarrow a^2 - b^2 = (a+b)(a-b)
 \end{aligned}
 \left| \begin{array}{l}
 x^2 + 8x + 16 \\
 a = x \quad b^2 = 16 \\
 \quad \quad \quad b = 4 \\
 2ab = 2 \times x \times 4 \\
 \rightarrow (x)^2 + 2(x)(4) + (4)^2 \\
 \quad \quad \quad (x+4)^2
 \end{array} \right.$$

$$\begin{aligned}
 &x^2 + 16 - 8x \\
 &a = x \quad b^2 = 16 \\
 &\quad \quad \quad b = 4 \\
 &2ab = 8x \\
 &\quad = 2 \times x \times 4 \\
 &(x)^2 + (4)^2 - 2(x)(4) \\
 &\quad \quad \quad (x-4)^2.
 \end{aligned}$$

$$4y^2 - 12y + 9$$

$$a^2 - 2ab + b^2$$

$$a^2 - 2ab + b^2 = (a-b)^2$$

$$a^2 + 2ab + b^2 = (a+b)^2$$

$$a^2 = 4y^2 \rightarrow 2ab = 12y \quad b^2 = 9 \quad (a-b)(a+b) = a^2 - b^2$$

square root $\rightarrow 2 \times 2y \times b = 12y$ $(3)^2 = 9$

$$\sqrt{a^2} = \sqrt{4y^2}$$

$$\rightarrow a = 2y$$

$$ab = 12y / 4y$$

$$b = 3$$

$$(a+b)^2$$

$$a^2 + 8a + 16$$

$$a^2 + 2ab + b^2$$

$$a = 2y \quad (2y-3)^2$$

$$b = 3$$

$$a^2 = a^2$$

$$a = a$$

$$b = 4$$

$$b^2 = 16$$

$$(4)^2 = 16$$

$$2ab = 8a$$

$$2 \times a \times b = 8a$$

$$b = \frac{8a}{2a}$$

$$\boxed{b = 4}$$

$$49x^2 - 36 = (7x - 6)(7x + 6)$$

$$\begin{aligned} & x^2 + 5x + 6 \\ & x^2 + 6x - x + 6 \\ & x^2 + 3x + 2x + 6 \\ & x(x+3) + 2(x+3) \\ & (x+3)(x+2) \end{aligned}$$

$$\boxed{2 \times 3} \quad 1 \times 6$$

$$5x = 6x - x$$

$$5x = 3x + 2x$$

$$x^2 + 5x + 6 = (x+3)(x+2)$$

$$25m^2 + 30m + 9$$

$$(5m + 3)^2$$

$$121b^2 - 88bc + 16c^2 = (11b - 4c)^2$$

$$m^4 - 100 = \boxed{(m^2 - 10)(m^2 + 10)} \quad a^2 = m^4 \quad b^2 = 100$$

$$a^2 - b^2 = (a - b)(a + b) \quad a = m^2 \quad b = 10$$

$$\text{Ans) } 4p^2 - 9q^2 = (2p - 3q)(2p + 3q)$$

$$y^2 - 7y + 12$$

$$y^2 - (3y + 4y) + 12$$

$$\cancel{y^2} - 3y - 4y + 12$$

$$\rightarrow y(y-3) - 4(y-3)$$

$$(y-3)(y-4)$$

$$3m(m+1) + 6(m+1)$$

$$(m+1)(3m+6)$$

$$3m^2 + \cancel{9m} + \cancel{6}$$

$$18 = 9 \times 2 \checkmark$$

$$= 3 \times 6 \checkmark$$

$$3m^2 + 3m + 6m + 6$$